A3-PS019-TC403

Personal-Data Discovery and Privacy Evidence Inventory

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Problem Description

Personal data piles up in unindexed tables and spreadsheets

- Organizations must know where PII is stored, why, and for how long
- Fields spread across CSV exports and SQL tables with no documentation
- Purpose and retention metadata is missing, stale or never linked to fields
- Reviewers need technical evidence, not automated legal verdicts

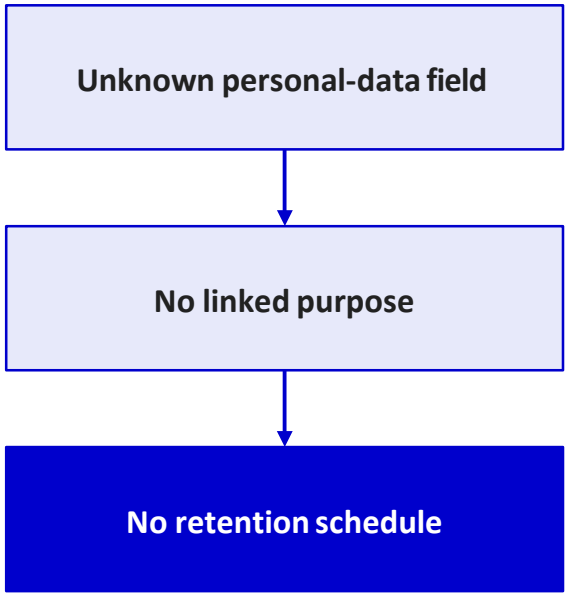
predefined data classes

- CSV + one SQL database
- Configurable detection rules
- Purpose and retention linking
- Missing-metadata flags

Background

Why a field-level evidence inventory is needed

- The DPDP Act, 2023 ties processing and storage of personal data to a declared purpose.
- DPDP Rules were notified in Nov 2025; erasure once the purpose ends is set out in Rule 8, with core duties phasing in about 18 months later.
- Privacy frameworks expect a PII inventory mapped to purposes.
- Without a field-level register, retention cannot be checked and gaps stay invisible.



Literature & RFC Review

Most relevant laws, standards and tools only

Reference	What it provides	Gap for this work
DPDP Act, 2023	Purpose-linked processing and storage limitation	No method to find where data sits
DPDP Rules, 2025	Erasure when purpose ends (Rule 8)	Needs purpose and retention per field
ISO/IEC 27701	PII management controls and records	Manual, document-driven
NIST Privacy Framework	Inventory and mapping outcomes	Outcome-level, no scanner
Microsoft Presidio	Open-source PII recognizers (regex, checksum, NER)	Text-oriented; no purpose or retention link

Proposed Solution

Five-stage scanner producing evidence for human review

1 Ingest	2 Detect	3 Link	4 Flag	5 Report
Read CSV files and one SQL database; profile columns and sample values	Configurable name and value rules with Luhn and Verhoeff check	Join fields to the purpose and retention register by dataset.field	Mark missing purpose or retention and low-evidence fields	Inventory, gap report and per-class precision / recall

Five data classes (proposed)

Name	Contact (email, phone)	Government ID	Financial account / card	Date of birth

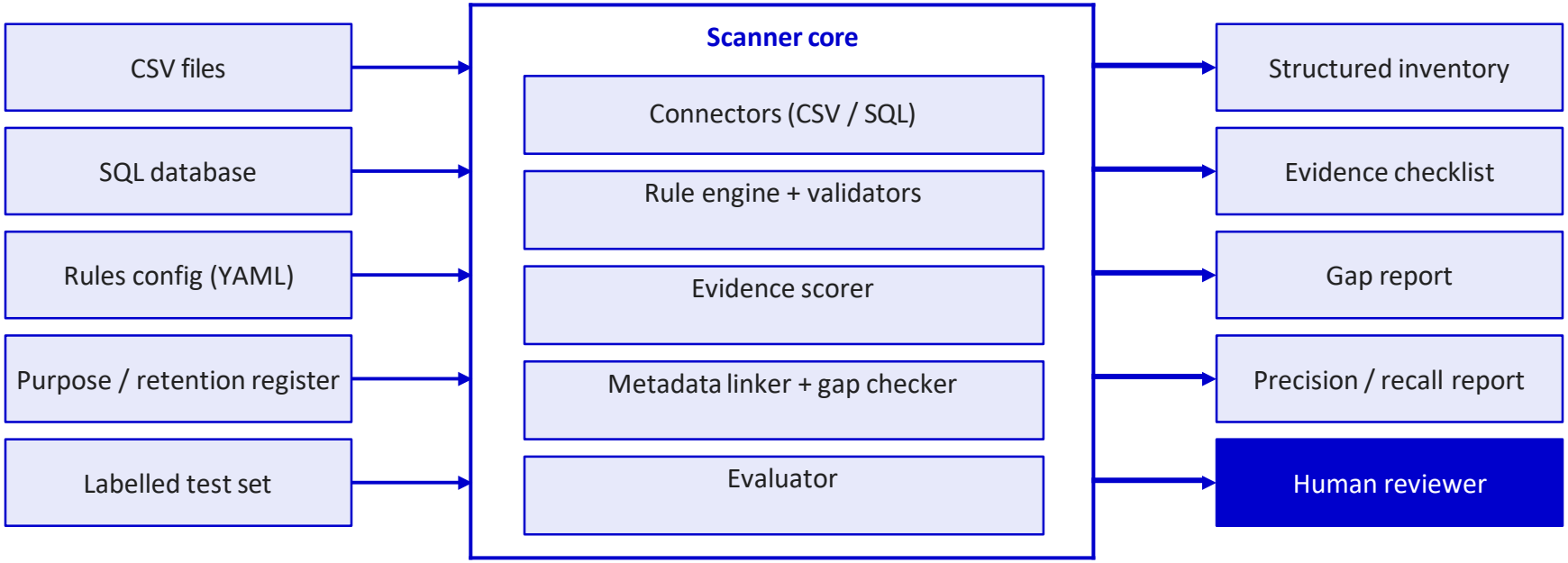
Optimization Proposed

Naive keyword scan versus evidence-based scan

Aspect	Before: name-only scan	After: proposed
Evidence	Column name match only	Name + value pattern + checksum evidence
False positives	Frequent, e.g. product_name	Confidence tiers; validators reject look-alikes
Scale	Full-table reads	Row sampling and per-column profiling
Configuration	Hard-coded patterns	YAML rules and evidence checklist
Report safety	Raw values in output	Masked samples only

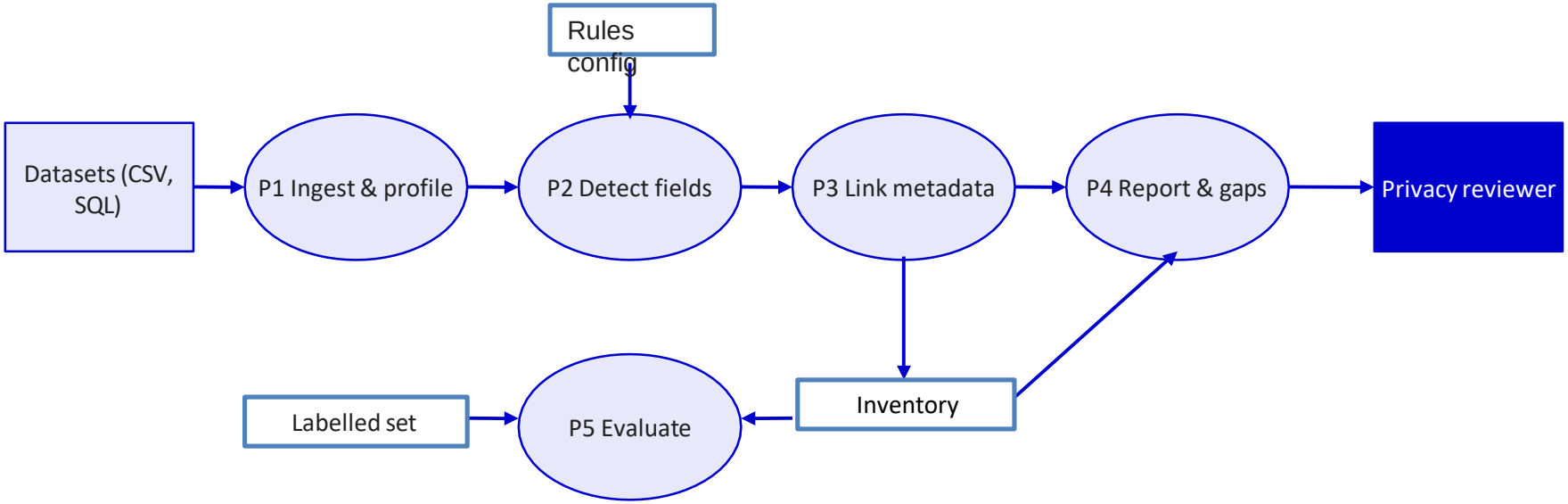
Solution Architecture & Design

Component view of the scanner



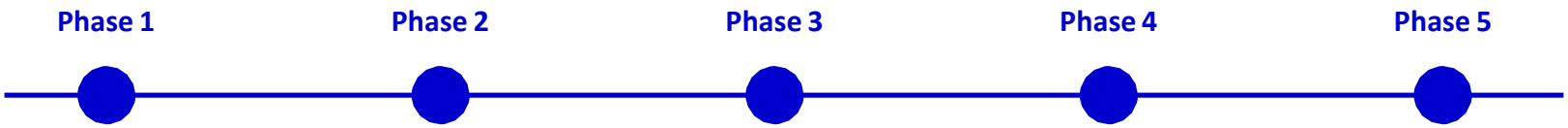
Data Flow Diagram

Level-1 flow from datasets to reviewed evidence



Deployment Timeline

Five phases, each ending in a runnable build for mentor review



Setup	Detect	Link	Evaluate	Deliver
Private GitHub repo, synthetic CSV + SQL data, labelled test set	Connectors, rule engine, validators	Purpose / retention register, gap flags	Per-class precision and recall, rule tuning	Technical report PDF

Methodology:

Setup, Detect, Link, Evaluate, and Deliver. Each phase involves implementation, testing, and evaluation, with performance improved through rule tuning and feedback. Every phase ends with a runnable build committed to GitHub for mentor review, ensuring continuous improvement and reliable delivery.

Bibliography

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Thank You for Your Attention!

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